



Parul University
Faculty of Engineering and Technology
Parul Institute of Engineering and Technology
Computer Science & Engineering Department

Subject Name	Cloud Computing	A.Y	2025-26
Subject Code	303105363	Semester	IV
Sr No	Question	COs	B.T
Unit 1: Introduction			
1	Define cloud computing and explain its essential characteristics.	CO2	Remember
2	Differentiate between public, private, and hybrid clouds.	CO3	Understand
3	Explain various layers of cloud architecture.	CO4	Apply
4	Discuss challenges in cloud computing adoption.	CO5	Analyze
5	Explain cloud infrastructure management with examples.	CO1	Evaluate
6	Write a note on applications of cloud computing.	CO2	Create
7	Explain service models used in cloud computing.	CO3	Remember
8	Discuss advantages and limitations of cloud computing.	CO4	Understand
9	What are cloud deployment models? Explain with examples.	CO5	Apply
10	Explain future trends in cloud computing.	CO1	Analyze
Unit 2: Virtualization			
Sr No	Question	CO	BT
1	Define virtualization and explain its need in cloud computing.	CO2	Remember
2	Differentiate between full virtualization and para-virtualization.	CO3	Understand
3	Explain storage virtualization with examples.	CO4	Apply
4	Discuss resource virtualization in cloud environments.	CO5	Analyze
5	Explain the working of a hypervisor.	CO1	Evaluate
6	What is virtual machine migration? Explain its benefits.	CO2	Create
7	Discuss the role of virtualization in cloud service delivery.	CO3	Remember
8	Explain network virtualization with suitable examples.	CO4	Understand
9	Compare virtualization and containerization.	CO5	Apply
10	Explain virtualization security challenges.	CO1	Analyze
Unit 3: Infrastructure as a Service (IaaS)			
Sr No	Question	CO	BT
1	Define IaaS and explain its architecture.	CO2	Remember
2	Explain the working of hypervisors in IaaS.	CO3	Understand
3	Discuss resource provisioning in IaaS environments.	CO4	Apply
4	Explain how IaaS is implemented in cloud platforms.	CO5	Analyze
5	Differentiate between IaaS and PaaS.	CO1	Evaluate

6	Write a note on IaaS service providers.	CO2	Create
7	Discuss advantages and challenges of IaaS.	CO3	Remember
8	Explain VM deployment process in IaaS.	CO4	Understand
9	Explain scalability in IaaS systems.	CO5	Apply
10	Explain cost models used in IaaS.	CO1	Analyze
Unit 4: Platform as a Service (PaaS)			
Sr No	Question	CO	BT
1	Define PaaS and explain its architecture.	CO2	Remember
2	Discuss cloud platform management with examples.	CO3	Understand
3	Explain steps to implement PaaS.	CO4	Apply
4	Differentiate between PaaS and SaaS.	CO5	Analyze
5	Explain runtime environment support in PaaS.	CO1	Evaluate
6	Discuss advantages and limitations of PaaS.	CO2	Create
7	Explain application development lifecycle in PaaS.	CO3	Remember
8	Explain integration capabilities of PaaS.	CO4	Understand
9	Discuss commercial PaaS offerings.	CO5	Apply
10	Explain PaaS deployment environment.	CO1	Analyze
Unit 5: Software as a Service (SaaS)			
Sr No	Question	CO	BT
1	Define SaaS and explain its characteristics.	CO2	Remember
2	Explain SaaS with Web 2.0 and Web OS examples.	CO3	Understand
3	Differentiate SaaS from traditional software delivery.	CO4	Apply
4	Discuss SaaS multi-tenancy architecture.	CO5	Analyze
5	Explain steps to implement SaaS.	CO1	Evaluate
6	Discuss SaaS pricing models.	CO2	Create
7	Explain SaaS application security challenges.	CO3	Remember
8	Write a note on data management in SaaS.	CO4	Understand
9	Discuss benefits of SaaS for enterprises.	CO5	Apply
10	Explain updates and maintenance in SaaS.	CO1	Analyze
Unit 6: Service Management in Cloud Computing			
Sr No	Question	CO	BT
1	Define service orchestration in cloud computing.	CO2	Remember
2	Explain service level agreements (SLAs) with examples.	CO3	Understand
3	Discuss billing and accounting models in cloud.	CO4	Apply
4	Explain economics of cloud scaling.	CO5	Analyze
5	Discuss data management challenges in cloud.	CO1	Evaluate
6	Explain cloud performance measurement parameters.	CO2	Create
7	Explain service monitoring in cloud environments.	CO3	Remember
8	Discuss capacity planning for cloud services.	CO4	Understand
9	Explain metering and monitoring in cloud.	CO5	Apply
10	Write a note on existing cloud project experiences.	CO1	Analyze

Unit 7: Security			
Sr No	Question	CO	BT
1	Explain cloud security requirements.	CO2	Remember
2	Discuss technologies used for data security in cloud computing.	CO3	Understand
3	Explain security concerns in cloud computing.	CO4	Apply
4	Discuss legal issues related to cloud environments.	CO5	Analyze
5	Explain cloud storage architecture from LAN to WAN.	CO1	Evaluate
6	Explain methods to secure private cloud.	CO2	Create
7	Explain methods to secure public cloud.	CO3	Remember
8	Discuss identity and access management (IAM) in cloud security.	CO4	Understand
9	Explain compliance challenges in cloud computing.	CO5	Apply
10	Discuss virtualization-related security risks.	CO1	Analyze
Unit 8: Case Studies			
Sr No	Question	CO	BT
1	Explain the architecture of Eucalyptus cloud.	CO2	Remember
2	Discuss services provided by VMware cloud.	CO3	Understand
3	Explain IBM Bluemix architecture and features.	CO4	Apply
4	Compare AWS and Google Cloud.	CO5	Analyze
5	Explain EC2 instance types and uses.	CO1	Evaluate
6	Discuss scalable storage solutions in AWS.	CO2	Create
7	Explain Google App Engine architecture.	CO3	Remember
8	Explain OpenStack components.	CO4	Understand
9	Discuss cloud orchestration tools.	CO5	Apply
10	Write a note on real-time cloud applications.	CO1	Analyze
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